Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. We trained each model for 100 epochs with a batch size of 64.

2 Results

Figure 1 shows the f1 score of CNN by dataset.

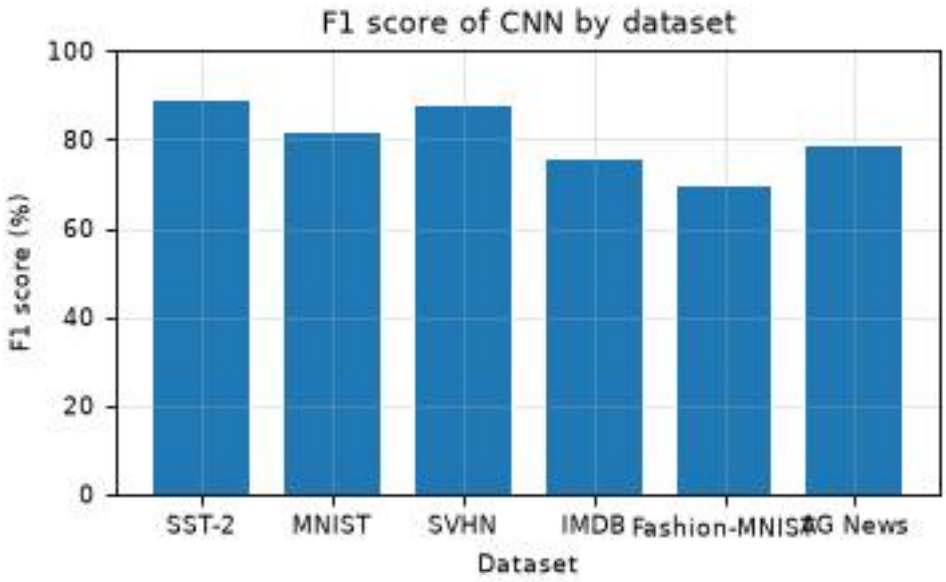


Figure 1: F1 score of CNN by dataset for the models in this experiment.

As shown in Figure 1, on AG News, CNN obtains a F1 score of 128.7%. CNN scores about 10 points higher on AG News than on Fashion-MNIST. CNN scores 13.2 points higher on SVHN than on Fashion-MNIST. CNN scores about 18 points higher on SVHN than on Fashion-MNIST (Figure 1).

Figure 2 shows the accuracy during training.

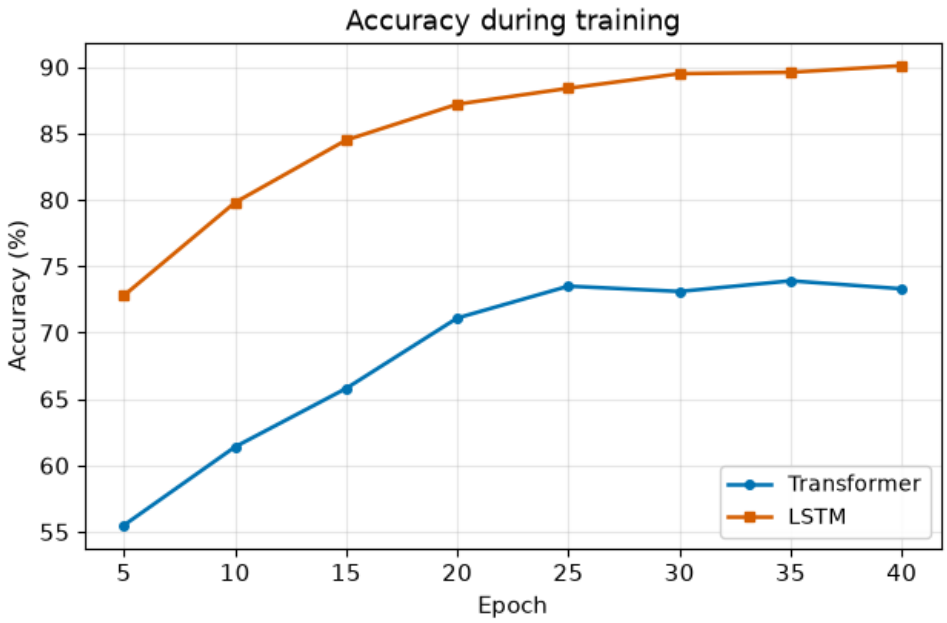


Figure 2: Accuracy during training for the models in this experiment.

LSTM outperforms Transformer by 18.4 percentage points at epoch 10. As shown in Figure 2, LSTM outperforms Transformer by 16.8 percentage points at epoch 40. LSTM outperforms Transformer by 20.6

percentage points at epoch 15.

3 Discussion

Logistic Regression reaches 90.4% accuracy on AG News. The differences between the models are discussed in the next section.